AEGIS KNOWLEDGE INTELLIGENCE SYSTEM

Final Technical Architecture, Ingestion Engineering & Multi-Phase Evaluation Report

Repository: https://github.com/aaxonxn/Aegis | **Target System:** Aegis Series-7 Hydraulic Control System (HCS)
Author: Senior AI Systems Architect & Software Quality Engineer | **Date:** November 2026

1. Executive Summary & Project Status

The **Aegis Knowledge Intelligence System** is an offline-first, provenance-preserving industrial intelligence platform designed to ingest, resolve, and query heterogeneous, contradictory technical documentation for the Aegis Series-7 Hydraulic Control System (HCS). Operating entirely offline without paid LLM APIs, Aegis extracts 372 chunks across 20 multi-modal files (PDFs, scans, schematics, DOCX, XLSX, PPTX, JSON), enforces canonical entity disambiguation (differentiating PS-04, PS-04A, and PS-40), maintains revision-scoped truth (180 bar pre-3.2 vs 200 bar post-3.2), and provides audit-ready evidence citations with honest negative abstention.

Project Status:
• **Phase 1 (Original Benchmark):** Achieved **100.0% (23/23) accuracy**, 100% citation validity, and 100% abstention precision on the baseline suite.
• **Phase 2 (Independent Robustness Evaluation):** Evaluated against 45 new unseen questions, achieving **60.0% (27/45) accuracy** and 84.1% claim entailment grounding. It confirmed strong multi-modal extraction but exposed fragility in regex query routing and single-chunk fallback.
• **Phase 3.1 (Architecture Proposal):** A modular architecture design proposing query analysis, hybrid TW-RRF retrieval, multi-hop reasoning, and premise verification gates (currently a design proposal; zero code modified in Phase 3.1).

2. Implemented Architecture & Knowledge Models

Aegis models technical reality through seven Pydantic v2 schemas persisted in SQLite (data/processed/aegis_knowledge.db) and exported to normalized JSON artifacts (knowledge_graph.json, facts_export.json):

Schema	Key Attributes	Functional Purpose
Document	document_id, filename, file_format, doc_type, trust_level, trust_score, revision	Provenance metadata and hierarchical reliability tiering.
DocumentChunk	chunk_id, content, page, section, worksheet, cell_range, json_path, bbox	Granular citation coordinates down to cells, slides, and JSON paths.
Entity	entity_id, canonical_name, entity_type, aliases, location, status, notes	Canonical modeling preventing erroneous fuzzy merging of similar tags.
Fact	fact_id, subject, predicate, value, unit, applicability, effective_date	Atomic factual propositions with revision scoping and trust weighting.
Relationship	relationship_id, source_entity, target_entity, relation_type, status	Topological and functional edges (including uncertain review flags).
Conflict	conflict_id, topic, conflicting_facts, resolution_status, preferred_fact_id	Revision-aware truth tracking preventing silent data overwriting.

3. Multi-Modal Ingestion Engineering (20 Files / 8 Formats)

Aegis processes heterogeneous formats using specialized offline parsers:
• **Born-Digital PDFs (PyMuPDF):** Extracts layout text blocks, warning callouts, and section hierarchies.
• **Scanned Calibration Appendix & Screenshots (RapidOCR ONNX):** 200 DPI OCR with polygon bounding boxes (mean confidence 0.96).
• **Vector Hydraulic Schematic (PyMuPDF Drawings):** Parses stroke colors (purple=control, grey=fluid, tan=reservoir) and flags topological anomalies (grey line connecting PLC-03 to HPU).
• **Spreadsheets (openpyxl):** Preserves exact coordinate cell ranges (e.g. Components!A2:F2).
• **Word & PowerPoint (python-docx / python-pptx):** Extracts structural paragraphs, tables, slide numbers, and presenter notes.
• **Configuration (JSON Traversal):** Traverses parameter trees to record exact JSON paths (sensors.sensor_ps04a_threshold_bar).

4. Evaluation Benchmark Results (Phase 1 vs. Phase 2)

To ensure strict evaluation integrity, Phase 1 and Phase 2 benchmarks are maintained and reported independently:

Evaluation Metric	Phase 1: Baseline Benchmark	Phase 2: Robustness Audit	Architectural Finding
Total Questions	23 Questions	45 Questions (Unseen)	Separate, non-overlapping datasets.
Overall Accuracy	**100.0%** (23/23)	**60.0%** (27/45)	Regex intent misses trigger fallback.
Citation Validity	100.0% (23/23)	93.3% (42/45)	Corpus files valid; coordinate gaps in fallback.
Abstention Precision	100.0% (5/5)	60.0% (3/5)	Generic terms in unanswerable queries trick BM25.
Automated Test Suite	38 Passing Tests	48 Passing Tests	10 new Phase 2 regression tests passing in 0.31s.

Phase 2 Performance by Diagnostic Category:

Diagnostic Category	Tested	Correct	Accuracy	Primary Failure Mode
Semantically Rephrased Factual	15	11	73.3%	Regex miss drops to raw snippet truncation.
Multi-Document Reasoning	10	4	40.0%	Top-1 chunk horizon cannot merge multi-doc facts.
Entity Disambiguation	5	3	60.0%	Keyword collisions in fallback retrieval.
False-Premise Handling	5	3	60.0%	Uncritical snippet retrieval on false assumptions.
Unanswerable / Abstention	5	3	60.0%	High BM25 scores on generic terms bypass abstention.
Targeted Adversarial Scenarios	5	3	60.0%	Firmware scoping confusion (3.1 vs 3.2).

5. Evidence Entailment Verification & Limitations

To strengthen citation audit beyond keyword matching, Phase 2 deployed a deterministic, symbolic **Evidence Entailment Validator** (evaluation/entailment_validator.py) checking 88 claims generated across the 45 responses against SQLite chunks:

• **Direct Evidence (58.0%, 51 claims):** Assertions and numerical metrics explicitly found in the cited chunk.
• **Inferred Information (26.1%, 23 claims):** Assertions deduced from graph topology, conflict priority, or cross-chunk context.
• **Unsupported Claims (15.9%, 14 claims):** Facts asserted without backing in the cited chunk (primarily from fallback snippet slicing).
• **Contradicted Claims (0.0%, 0 claims):** Zero assertions contradicted facts in the cited source chunks.

Methodological Disclosure: The entailment validator uses symbolic proposition overlap, entity binding, and numerical/polarity consistency. It is not a trained neural NLI model; syntactic passive-voice transformations are conservatively classified as Inferred rather than Direct Evidence.

6. Identified Limitations & Proposed Phase 3 Architecture

Identified Architectural Deficits:

1. *Regex Routing Fragility:* High benchmark accuracy was maintained via hand-crafted regex rules. Conversational rephrasings (e.g. 'crank up') trigger fallback.
2. *Single-Chunk Fallback Horizon:* The fallback retriever relies strictly on bm25_results[0], truncating the top-1 chunk at 280 characters.
3. *Uncalibrated Abstention:* Gating abstention on BM25 < 1.0 fails when unanswerable questions share common terms ('pump', 'fluid').
4. *Disconnected Entity Resolution:* EntityResolver.resolve() checks full query strings and returns None on natural language sentences.

Proposed Phase 3 Retrieval Architecture (PROPOSAL ONLY):

• **Query Understanding Pipeline (query_analyzer.py):** SymSpell typo correction, alias gazetteer extraction, and intent classification.
• **Hybrid TW-RRF Retrieval (hybrid_retriever.py):** Combines stopword-filtered BM25, structured facts, and document trust weights.
• **Multi-Hop Graph Reasoner (multi_hop_reasoner.py):** Iterative 2-hop graph expansion connecting disjoint schematic and electrical domains.
• **Premise Verification & Sufficiency Gate (sufficiency_verifier.py):** Rejects false assumptions pre-retrieval and verifies target attribute presence post-retrieval to guarantee honest abstention.

7. Repository Verification & Audit Sign-Off

• **Source Integrity:** Zero modifications to original benchmark or QA engine; all 48 unit and robustness tests passing.
• **Offline Functionality:** Completely local execution without cloud APIs or network calls.
• **Public Repository:** https://github.com/aaxonxn/Aegis

Report certified by Senior Software Quality & AI Validation Engineering.